

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: Data Analysis with SAS / SPSS / R

PRACTICAL 12

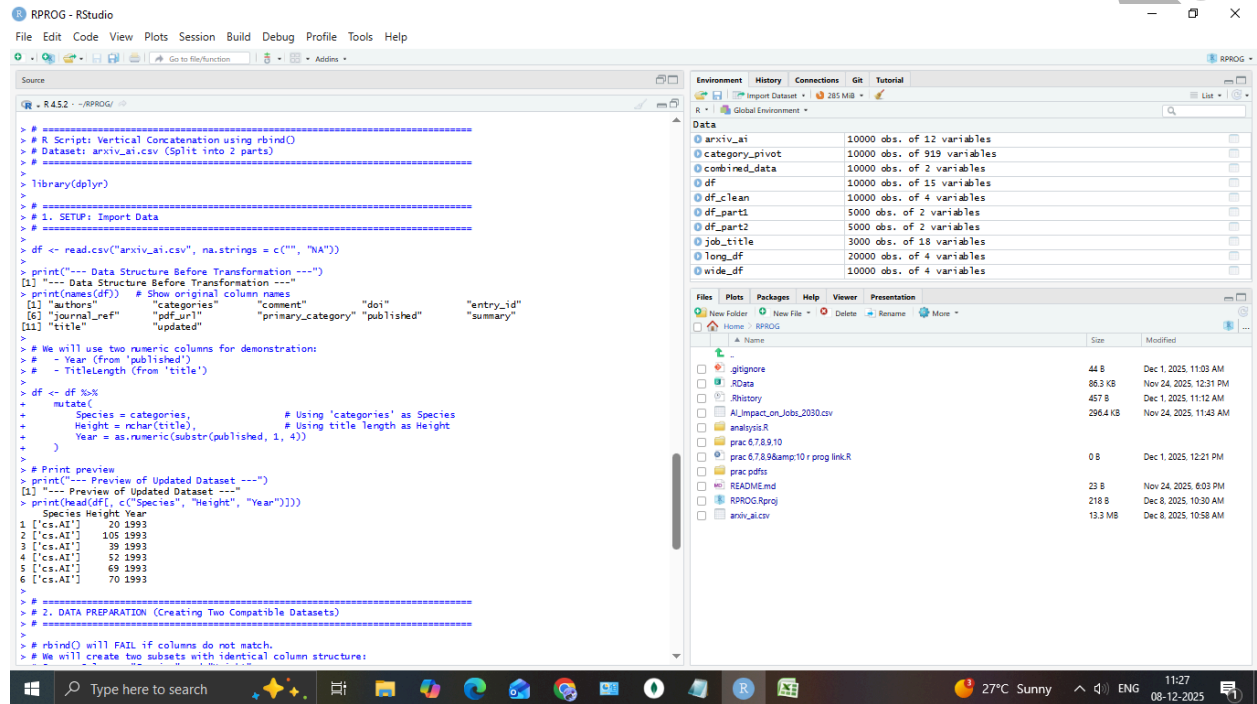
AIM: 12 Combining datasets vertically (concatenation) using rbind() (R).

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```
RPROG - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Source
R 4.5.2 - ~/RPROG/
> # =====
> # 2. DATA PREPARATION (Creating Two Compatible Datasets)
> # =====
> # rbind() will FAIL if columns do not match.
> # We will create two subsets with identical column structure:
> # Common Columns: "Species" and "Height"
>
> # 2.1 First half of the dataset
> df_part1 <- df[floor(nrow(df)/2):nrow(df), c("Species", "Height")]
> names(df_part1) <- c("Species", "Height")
>
> # 2.2 Second half of the dataset
> df_part2 <- df[floor(nrow(df)/2+1):nrow(df), c("Species", "Height")]
> names(df_part2) <- c("Species", "Height")
>
> # Ensure both Height columns are numeric
> df_part1$Height <- as.numeric(df_part1$Height)
> df_part2$Height <- as.numeric(df_part2$Height)
>
> # =====
> # 3. VERTICAL COMBINATION (rbind)
> # =====
> combined_data <- rbind(df_part1, df_part2)
>
> print("---- Combined Data Summary ----")
> print(paste("Part 1 rows:", nrow(df_part1)))
> print(paste("Part 2 rows:", nrow(df_part2)))
> print(paste("Total rows (Expected):", nrow(df_part1) + nrow(df_part2)))
> print(paste("Total rows (Expected): 10000"))
> print(paste("Total rows (Actual):", nrow(combined_data)))
> print(paste("Total rows (Actual): 10000"))
>
> print("---- Preview of Combined Data (Top and Bottom) ----")
> print(head(combined_data)) # First few rows from part 1
  Species Height
1 ['cs.AI']    20
2 ['cs.AI']   105
3 ['cs.AI']    39
4 ['cs.AI']    52
5 ['cs.AI']    69
6 ['cs.AI']    70
> print(tail(combined_data)) # Last few rows from part 2
9995
      Species Height
9995 ['cs.AI', 'cs.LG'] 116
```

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```
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Source
R 4.5.2 -- RPROG/
> # 2.1 First half of the dataset
> df_part1 <- df[1:floor(nrow(df)/2), c("Species", "Height")]
> names(df_part1) <- c("Species", "Height")
>
> # 2.2 Second half of the dataset
> df_part2 <- df[(floor(nrow(df)/2)+1):nrow(df), c("Species", "Height")]
> names(df_part2) <- c("Species", "Height")
>
> # Ensure both Height columns are numeric
> df_part1$Height <- as.numeric(df_part1$Height)
> df_part2$Height <- as.numeric(df_part2$Height)
>
> # 3. VERTICAL COMBINATION (rbind)
>
> combined_data <- rbind(df_part1, df_part2)
>
> print("---- Combined Data Summary ----")
[1] "---- Combined Data Summary ----"
> print(paste("Part 1 rows:", nrow(df_part1)))
[1] "Part 1 rows: 5000"
> print(paste("Part 2 rows:", nrow(df_part2)))
[1] "Part 2 rows: 5000"
> print(paste("Total rows (Expected):", nrow(df_part1) + nrow(df_part2)))
[1] "Total rows (Expected): 10000"
> print(paste("Total rows (Actual):", nrow(combined_data)))
[1] "Total rows (Actual): 10000"
>
> print("---- Preview of Combined Data (Top and Bottom) ----")
[1] "---- Preview of Combined Data (Top and Bottom) ----"
> print(head(combined_data)) # First few rows from part 1
  Species Height
1 ['cs.AI']    20
2 ['cs.AI']   105
3 ['cs.AI']    39
4 ['cs.AI']    52
5 ['cs.AI']    69
6 ['cs.AI']    70
> print(tail(combined_data)) # Last few rows from part 2
      Species Height
9995 ['cs.AI', 'cs.L0'] 116
9996 ['cs.CL', 'cs.AI'] 102
9997 ['q-fin.RM', 'cs.AI'] 49
9998 ['cs.NI', 'cs.AI'] 78
9999 ['cs.L0', 'cs.AI'] 63
10000 ['cs.AI', 'cs.L0', '03860, 03815, 68T27, 68T30, 68T15', 'I.2.3; I.2.4; I.2.0; F.4.1'] 50
>
```

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